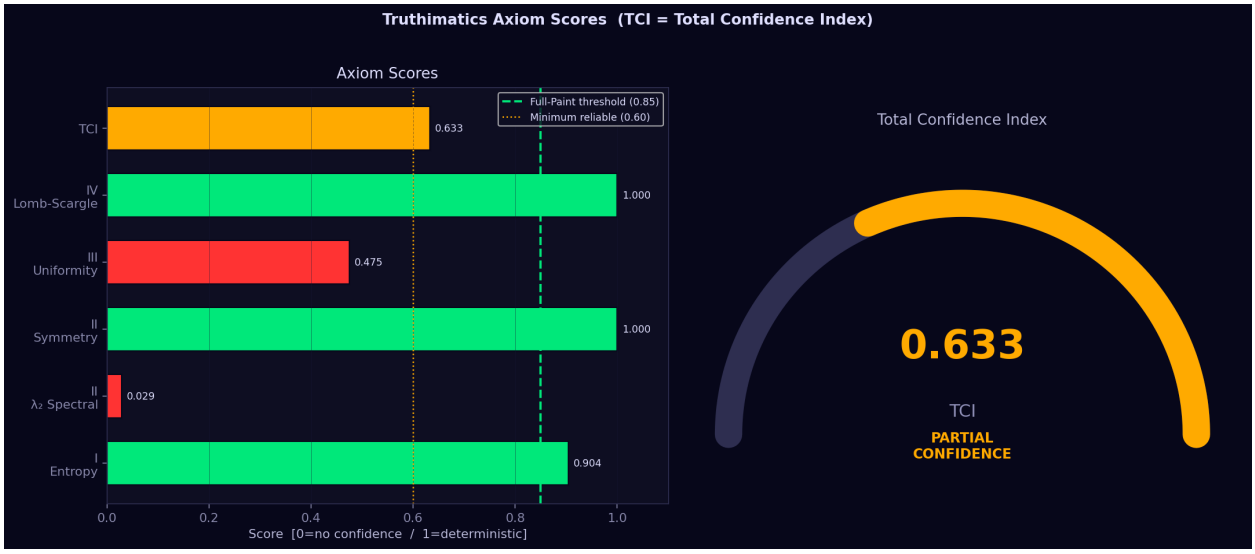
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_032682C9D5
Taxa	Quetzalcoatlus
Clade	Pterosaur
PBDB Records	8
Mass (regression)	218.8 kg (Campione & Evans 2012)
Mass (volumetric)	56.3 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	0.072 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6327 (partial)
Axiom I Entropy	0.9041
Axiom II λ■	0.1171 (symmetry 100%)
Axiom III Uniformity	0.4746
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	66.7% Measured / 33.3% Inferred
Elements	24
CoG (m)	[0.464, 0.339, 0.0]
Generated	2026-05-01 00:03 UTC

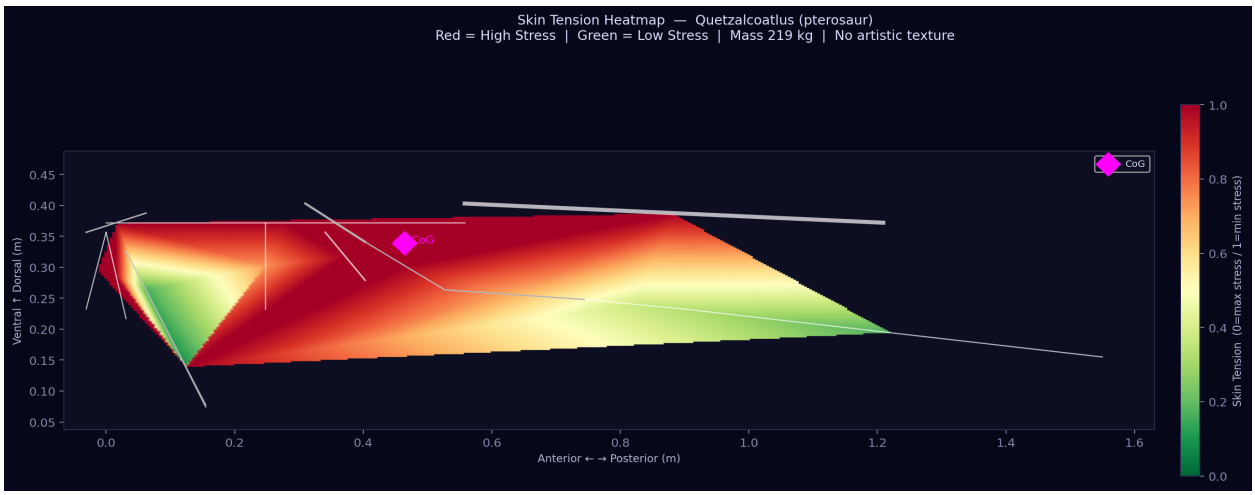
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



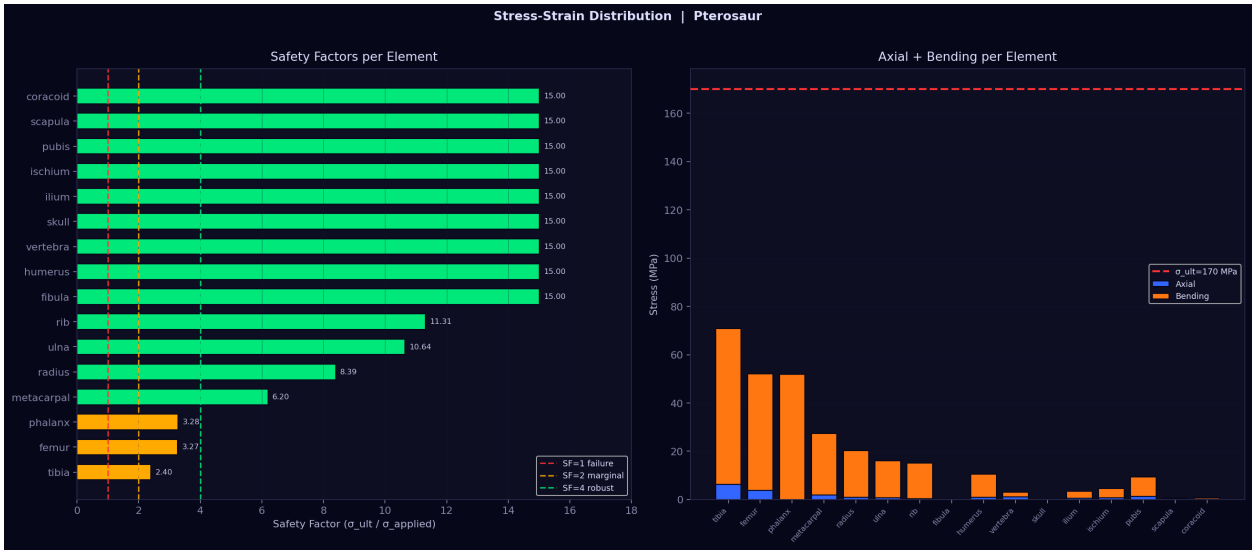
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

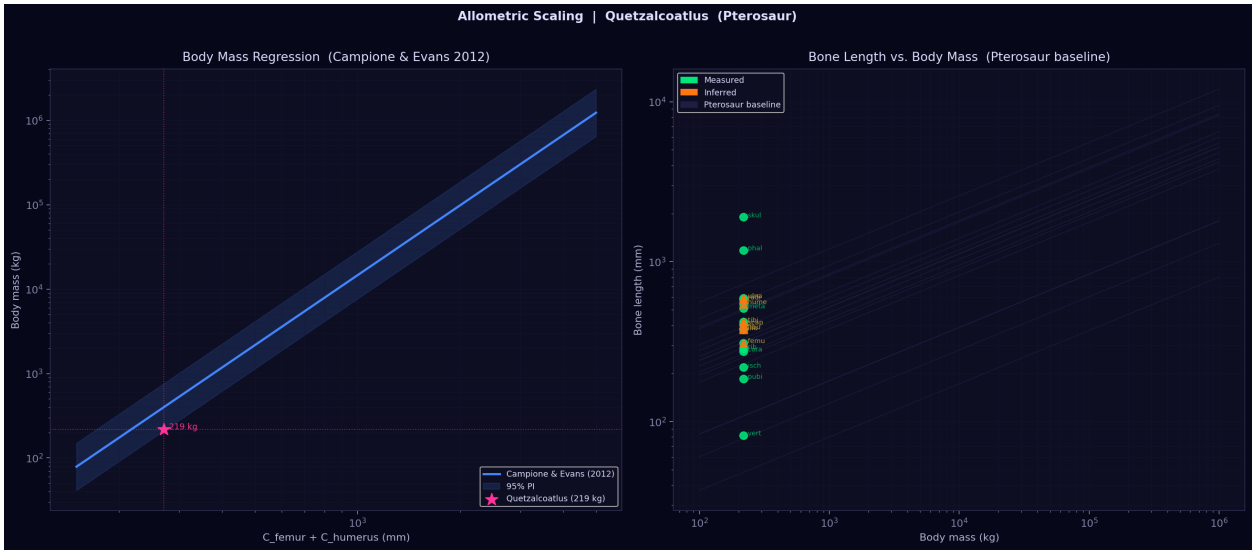


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.158	0.596	0.754	15.000	OK
femur	3.863	48.205	52.068	3.265	WARNING
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.900	9.517	10.418	15.000	OK
ilium	0.584	2.744	3.328	15.000	OK
ischium	0.734	3.797	4.531	15.000	OK
metacarpal	1.945	25.488	27.432	6.197	OK
phalanx	0.000	51.799	51.799	3.282	WARNING
pubis	1.352	8.055	9.407	15.000	OK
radius	0.973	19.282	20.256	8.393	OK
rib	0.417	14.618	15.036	11.306	OK
scapula	0.091	0.384	0.475	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	6.257	64.629	70.886	2.398	WARNING
ulna	0.818	15.160	15.977	10.640	OK
vertebra	1.120	1.991	3.111	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	0.839 m	Femur + tibia + metatarsal
Estimated Speed	5.5 km/h (1.52 m/s)	Gait: Walk | Fr=0.28
Stride Length	1.314 m	Alexander (1976) / Thulborn (1990)
Bite Force	2057.1 N (209.7 kgf)	Pterosaur regression (Witton 2010)
Skull Length	1900.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.2838	Circumference / length (dimensionless)
Limb Strength Index	1.4e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	2.79 m	Allometric from femur length
Est. Neck Length	0.84 m	Allometric from femur length
Est. Tail Length	0.22 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	275.0	82.0	740.66	924.6	Measured	0.88	15.00
femur	310.0	28.0	97.35	121.5	Measured	0.88	3.27
femur_right	310.0	28.0	97.35	121.5	Symmetry	0.84	3.27
fibula	380.0	8.0	6.88	7.2	Measured	0.88	15.00
fibula_right	380.0	8.0	6.88	7.2	Symmetry	0.84	15.00
humerus	544.0	58.0	733.02	915.0	Measured	0.88	15.00
humerus_right	544.0	58.0	733.02	915.0	Symmetry	0.84	15.00
ilium	375.0	72.0	778.68	972.0	Measured	0.88	15.00
ilium_right	375.0	72.0	778.68	972.0	Symmetry	0.84	15.00
ischium	218.0	38.0	126.09	157.4	Measured	0.88	15.00
metacarpal	510.0	43.0	140.72	115.8	Measured	0.88	6.20
phalanx	1180.0	28.0	111.60	86.4	Measured	0.88	3.28
pubis	185.0	28.0	58.10	72.5	Measured	0.88	15.00
radius	580.0	33.0	253.00	315.8	Measured	0.88	8.39
radius_right	580.0	33.0	253.00	315.8	Symmetry	0.84	8.39
rib	285.0	11.0	13.81	17.2	Measured	0.88	11.31
scapula	405.0	108.0	1892.18	2362.0	Measured	0.88	15.00
scapula_right	405.0	108.0	1892.18	2362.0	Symmetry	0.84	15.00
skull	1900.0	210.0	33562.34	41895.9	Measured	0.88	15.00
tibia	420.0	22.0	81.42	101.6	Measured	0.88	2.40
tibia_right	420.0	22.0	81.42	101.6	Symmetry	0.84	2.40
ulna	592.0	36.0	307.32	383.6	Measured	0.88	10.64
ulna_right	592.0	36.0	307.32	383.6	Symmetry	0.84	10.64
vertebra	82.0	52.0	88.81	110.9	Measured	0.88	15.00